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Spermidine-induced two-dimensional DNA condensations on mica surfaces: A different pathway from condensations in solution

ZHANG XingHua, HOU XiMiao, JI Chao, LI Ming, DOU ShuoXing & WANG PengYe†

Laboratory of Soft Matter Physics, Beijing National Laboratory for Condensed Matter Physics, Institute of Physics, Chinese Academy of Sciences, Beijing 100190, China

With atomic force microscopy (AFM) we systematically studied the DNA condensations on mica surfaces induced by multivalent cation spermidine. The pattern of the DNA condensates is a flat single layer, with a core in the centre and DNA wrapping around it in high density. We assume this to be a two-dimensional condensation of free coiled DNA onto negatively charged mica surfaces by the multivalent cation. The DNA molecules condense on mica surfaces via a pathway different from the formation of toroids, rods or globules in bulk solutions. We give an explanation to why toroid structures are difficult to be observed by AFM, and further discuss the relationship between DNA condensations in solutions and on mica surfaces. The present work will be helpful for understanding the behaviors of DNA on charged surfaces, which might be significantly different from that in solutions.

DNA condensation, spermidine, two-dimension, condensing pathway, AFM

BIOPHYSICS

The condensation of DNA has become a fascinating research subject. Studies on DNA condensations are beneficial to the understanding of the physical behaviors of charged biopolymers and the process of virus packing *in vivo*, as well as to the development of technologies for gene therapy[1,2]. Experimental evidences have shown that DNA condensation occurs when ~90% of its charge is neutralized by multivalent cations, leading to a weak short range attractive force between different DNA segments[3]. DNA condensing agents include spermidine, spermine, cobalt hexamine and proteins such as protamine. The condensation can also be induced by crowding polymers such as PEG or weak electrolyte such as ethanol[4,5]. Although appearing as being continuous in ensemble assays, the coil-globule transition of DNA is markedly discrete for individual molecules, which is considered as a first-order phase transition[6]. In other words, each DNA molecule exists either in coiled or folded states. Fluorescence microscopy observations have revealed that in bulk solutions, DNA molecules are mainly elongated when the spermidine concentration is less than 100 μM; elongated coils and folded compact states coexist in 150 μM spermidine; and all DNA molecules are in folded state when the spermidine concentration is increased to 200 μM[7].

Toroid is the most frequently observed folded structure of DNA condensates in solutions[8]. The size of the toroids is largely independent of the DNA length, reflecting that several DNA molecules may be incorporated into a single particle. Toroids are usually observed with electron microscopy. In such an observation, DNA is condensed in solution in advance, and then deposited onto a hydrophobic carbon grid and fixed with ethanol[9]. Some very well developed toroids have been recorded using this method.

As a convenient and high-resolution instrument, AFM

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†Corresponding author (email: pywang@aphy.iphy.ac.cn)
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is also often used to study the DNA condensates formed on solid surfaces[10,11]. The experiments have one thing in common, that is, the employment of mica surfaces. Protamine was observed to be able to condense DNA into toroid structures on mica surfaces[12]. In this experiment, DNA was bound to the surface loosely with Mg^{2+} in advance before protamine was added to the solution of the pre-bound DNA. Similarly, when DNA was bound loosely with Mg^{2+} to mica, rinsing the surface with ethanol could also lead to the formation of toroids[13]. In addition, silanes could condense DNA into toroids on silicon or mica surfaces[14].

However, when multivalent cations such as spermidine or hexamine cobalt were used, well-defined structures like toroids or rods could hardly be observed on the mica surfaces, except for flat patterns of DNA. Fang and Hoh[15] reported the flowerlike patterns of spermidine-induced DNA condensation on mica surfaces which were interpreted as early intermediates of free toroids in solutions. In 2007, the influence of charged surface on morphologies of spermidine-induced DNA condensates was investigated systemically[16]. Again, flowerlike shapes of a single layer were formed on mica surfaces. The authors ascribed these morphologies to the rearrangement of DNA, proposing that the negatively charged mica surfaces disturbed the three-dimensional toroids formed in solutions. However such an idea cannot explain why protamine can condense DNA into toroids on mica surfaces whereas spermidine cannot. Besides, one would wonder how spermidine condenses DNA into such flat flowerlike patterns rather than toroids on the mica surfaces. Are they generated by the same or different pathways[17]? Actually, besides flat patterns, toroid structures were observed using the similar method of condensing DNA in solution and then transferring condensed DNA onto mica surface[18]. The difference was that the sample was dried by heating there, rather than nitrogen stream.

In order to understand the mechanism of DNA condensations on mica surfaces, we used different methods to condense DNA. Our results indicate that the spermidine-induced condensation of DNA on mica surfaces may follow a different pathway from the DNA condensation to toroids in bulk solutions. On mica surfaces, it is a two-dimensional condensation of coiled DNA. The multivalent cations generally contribute in two ways to the condensation: (1) attaching the negatively charged DNA to the negatively charged mica surface; (2) neutralizing the charges of DNA itself and thus inducing strand-strand attraction which leads to the well-ordered DNA patterns. Thus under the concerted influences of the DNA-mica adsorption and DNA-DNA attraction, two-dimensional DNA condensations occur on negatively charged mica surface.

1 Materials and methods

1.1 DNA and chemicals

λ-DNA (48502 bp) was purchased from Sino-America Biotechnology Company in Shanghai. All DNA we used in the experiments were dissolved in 10 mM Tris-HCl buffer with pH 7.5. Other chemicals were all purchased from Sigma-Aldrich. Spermidine ($[C_7N_3H_{22}]^{3+}$) was purchased as chloride salts and dissolved in pure water before use. Protamine was dissolved in buffer (50 mM Tris-HCl, pH 7.8, 120 mM KCl, 10 mM $MgCl_2$, 0.5 mM DTT and 0.2 μg/μL BSA).

1.2 Sample preparation

(i) DNA adsorbed on mica surface by divalent cations. Samples were prepared by depositing a 10 μL droplet of 1 ng/μL λ-DNA in 10 mM Tris-HCl (pH 7.5) and 5 mM Mg^{2+} or 1 mM Ni^{2+} onto freshly cleaved mica. After 5 min, mica surface was washed with 200 μL Milli-Q filtered water for several times and blown dry in a gentle stream of nitrogen gas.

(ii) DNA-spermidine complexes on mica substrates were prepared by using three different methods

Method 1 Equal volumes of λ-DNA solution (2 ng/μL concentration) and spermidine solution (100 μM to 10 mM concentration) were mixed in advance in a tube. After incubation at room temperature for 30 min, a 10 μL aliquot of the mixture was deposited onto a freshly cleaved mica surface, and incubated for 5 min. The samples were then gently rinsed with 200 μL pure water, and completely blown dry with a stream of nitrogen gas.

Method 2 Aliquots of 10 μL λ-DNA solution (2 ng/μL concentration) and 10 μL spermidine solution of different concentrations were deposited simultaneously onto a new cleaved mica surface, and mixed slowly with a pipette. After incubation for 30 min, the mica surface was gently rinsed with 200 μL pure water, and completely blown dry with a stream of nitrogen gas.

Method 3 An aliquot of 10 μL spermidine solution of various concentrations was deposited onto a freshly cleaved mica surface at first. After 10 min, 10 μL λ-DNA solution (2 ng/μL concentration) was added to the pretreated surface. 10 min later, the surface was rinsed and dried in the same way as described above.

(iii) DNA-protamine complexes on mica substrates were prepared using three different methods.

Method 4 Equal volumes of λ-DNA in 10 mM Tris-HCl (pH 7.5) (1 ng/μL concentration) and protamine diluted in pure water (1 ng/μL concentration) were mixed in tube. After incubation at room temperature for 30 min, Ni^{2+} (or Mg^{2+}) with a final concentration of 1 (or 5) mM was added to the solution, followed by depositing 10 μL aliquots of the solution onto a freshly cleaved mica surface. After 5 min, the mica surface was gently rinsed with 200 μL pure water, and completely blown dry with a stream of nitrogen gas.

Method 5 Aliquots of 10 μL λ-DNA (1 ng/μL concentration) in 10 mM Tris-HCl with 1 mM Ni^{2+} and 10 μL protamine diluted in pure water (1 ng/μL concentration) were deposited simultaneously onto a new cleaved mica surface, and mixed slowly with a pipette. After incubation for 10 min, the surface was gently rinsed with 200 μL pure water, and completely blown dry with a stream of nitrogen gas.

Method 6 An aliquot of 10 μL λ-DNA (1 ng/μL concentration) in 10 mM Tris-HCl (pH 7.5) with 2.5 or 10 mM Mg^{2+} was deposited onto a freshly cleaved mica surface for 2 min, then an aliquot of 10 μL protamine diluted in pure water (1 ng/μL concentration) was added, and incubated for 2 min. The surface was then rinsed and dried as above.

1.3 AP-mica preparation

50 μL APTES (aminopropyltriethoxysilane) was deposited into a little cap. Afterwards, mica was placed onto the cap, with the freshly cleaved surface facing the liquid. The cap and mica were kept in a vacuum desiccator. After 1 h, the mica was taken out, and its surface was immediately blown dry with nitrogen stream. AP-mica was then ready for use. The DNA concentration in all AP-mica experiments was 0.1 ng/μL.

1.4 AFM imaging

The imaging was performed in air with a multi-mode AFM with nanoscope IIIa controller (Digital Instruments, Santa Barbara, CA, USA) in the tapping-mode. Silicon probe RTESP14 from Veeco (America) was employed, with a resonance frequency of 315 kHz. The 'E' scanner was used. The scan frequency was 1 Hz per line, and the scan size was between 1 and 5 μm.

2 Results

2.1 DNA condensation by spermidine

As a control, we first observed λ-DNA on mica surface adsorbed by divalent cation Mg^{2+} and Ni^{2+} (Figure 1). DNA all takes the form of random coils without condensation.

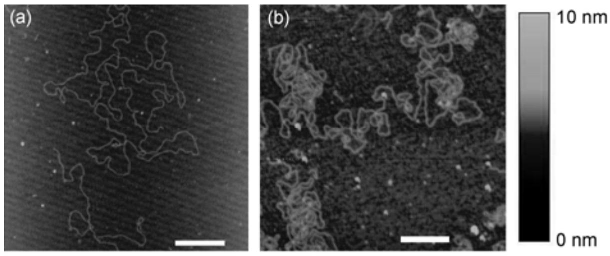

Figure 1 λ-DNA attached onto mica surface by divalent cations. (a) DNA looks extended and relaxed when adsorbed on mica by Mg^{2+}; (b) DNA exhibits as random coils but a little compact when adsorbed by Ni^{2+}. Scale bars are both 500 nm.

(i) Condensation patterns formed by pre-incubation. Samples prepared using method 1 by pre-mixing DNA and spermidine in tube were imaged and presented in Figure 2. Figure 2(a)—(d) represent the morphologies of DNA on mica surfaces with the spermidine concentrations of 100, 140, 250 and 300 μM respectively. In fact, even 350 μM spermidine could produce similar patterns in our experiments, though less frequently. It is apparent that the DNA segments are captured by the mica surface, forming structures with a circular profile. For each condensate, there is a core in the centre which is much higher than the surrounding DNA segments. DNA segments wrap around the core to form a flat disk, or in some circumstances, wrap in a way like the petals around the core of a flower (Figure 2(d)). The DNA segments are densely packed in these condensates, and sometimes it is even difficult to distinguish between the neighboring segments (Figure 2(c)). Most of these condensates are single layered, containing more than one DNA molecules, but still rather flat. Besides, the sizes are comparatively conserved, about 1 μm, despite that the spermidine concentrations range from 100 to 350 μM. When the spermidine concentration increases to 400 μM, the results become remarkably different: hardly any pattern can be found (Figure 2(e)). The results are

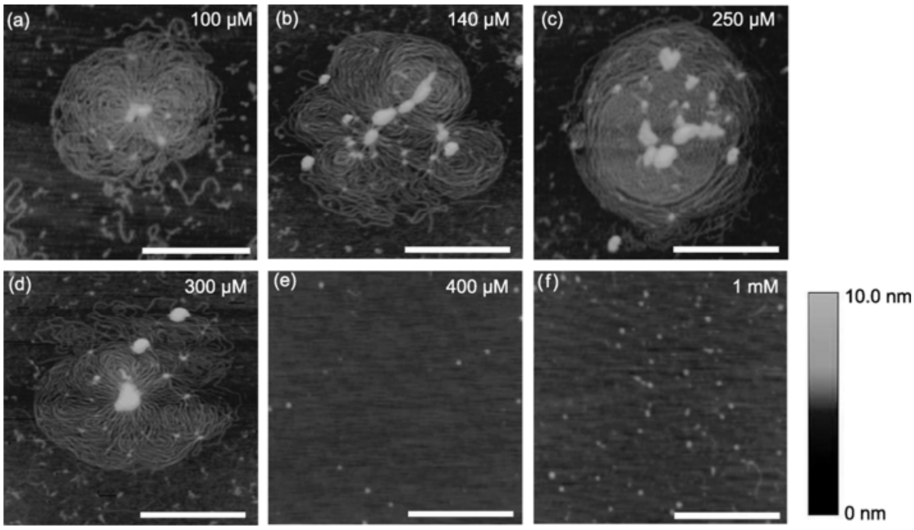

Figure 2 AFM images in air of DNA condensates prepared with different concentrations of spermidine on bare mica surface. Method 1 was employed. (a)—(d) show respectively the condensation patterns under 100, 140, 250 and 300 μM spermidine. (e) and (f) correspond to that when the spermidine concentration increases to 400 μM or higher. All scale bars are 500 nm.

similar when the spermidine concentration is further increased to 1 mM, and even to 5 mM. Such results are astonishing, probably indicating that something important happens with the increase of the spermidine concentration but cannot be observed directly on the mica surfaces.

In fact, from the following experiments, we may deduce what actually happens, which is presented here in advance (Figure 3). In the bulk solution, elongated DNA coils and folded toroids coexist, with the ratio of their numbers depending on the spermidine concentration. However, coiled DNA formed in the solution may be attached by spermidine onto mica surfaces and form flat patterns; while for the compact toroids formed in the solution, it is difficult for them to adhere to the charged surfaces, and to sustain the rinsing and nitrogen blowing. Accordingly, with the increase of spermidine concentration, less and less DNA molecules exist as coils in solution and thus less and less flat patterns can be observed on mica surfaces. This well explains why toroids are often observed in electron microscopy experiments[9] whereas only flat patterns can be observed in AFM experiments[15,16].

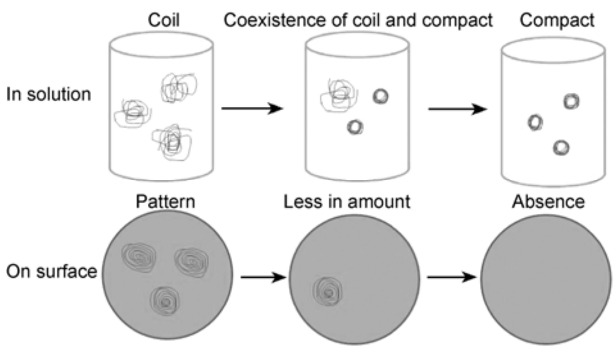

Figure 3 Sketch of the formation of DNA patterns on mica surfaces with Method 1.

(ii) Condensation patterns formed by direct incubation on mica surface. To check our assumption, we prepared the samples using method 2, namely, depositing the DNA and the spermidine solutions simultaneously onto a newly cleaved mica surface, and directly incubating there. Half an hour later, the samples were rinsed and blown dry. The morphologies are presented in Figure 4. With 100 μM spermidine, the DNA condenses into the same morphology as in method 1 (incubation in advance in tube). This result seems a little trivial. But with 1 mM spermidine, the DNA condenses into a flat and comparatively well-defined morphology like that shown in Figures 2(a)—(d). This is in contrast with the one observed in Figure 2(f) where no condensates could be formed on negatively charged mica surfaces when DNA and 1 mM spermidine were pre-incubated in tube. The difference can be explained by the kinetic effect: when directly mixed on mica surfaces, coiled DNA molecules are attracted onto the surface before they completely condense into three-dimensional compact globules in the solution. The observations therefore support our assumption that the two-dimensional patterns are due to the two-dimensional condensation of coiled DNA.

(iii) Condensation patterns on spermidine-modified

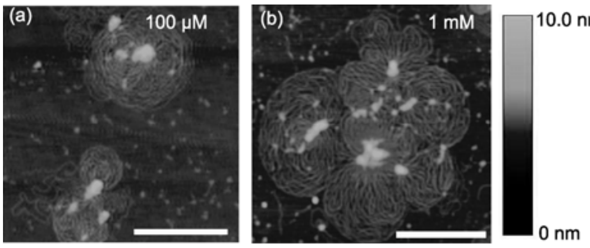

Figure 4 DNA condensates prepared with different concentrations of spermidine on bare mica surfaces using method 2. (a) 100 μM spermidine; (b) 1 mM spermidine. All scale bars are 500 nm.

mica surfaces. We further investigated the role played by spermidine in DNA condensations on mica surfaces using method 3, i.e. pre-treating the mica surface with various concentrations of spermidine for 10 min, then depositing DNA solution onto the surface. After incubation for 10 min, the samples were rinsed and blown dry in the same way as in the other two methods. The DNA morphologies are presented in Figure 5. With 100 μM spermidine, DNA is strongly adsorbed onto the mica surface, but in a disordered manner. We see that DNA covers most of the area on mica surface in a rather high density. This phenomenon reflects that spermidine is able to effectively modify mica surface with positive charges and consequently, DNA molecules are trapped promptly by the modified surface after DNA deposition. However, no pattern as that in Figures 2(a)—(d) is observed with this method. The DNA molecules exist instead as random coils on the spermidine modified mica surface. When the spermidine concentration is increased to 1 mM, DNA is also adhered to the mica surface densely as in the case of 100 μM. But in some regions, three-dimensional cores appear, and around the cores, DNA is organized as two-dimensional condensation, though less ordered (Figure 5(b)). It should be noted here that when we added DNA to the mica surface, the spermidine solution deposited in advance was not dried or adsorbed away, so that free spermidine molecules still existed in solution after the DNA addition. As a result, DNA condensation also occurred to some extent in the remaining spermidine solution, especially at high spermidine concentration, leading to the formation of three-dimensional cores. From the above analysis, we could easily see that spermidine can induce both DNA-mica adsorption and DNA-DNA attraction.

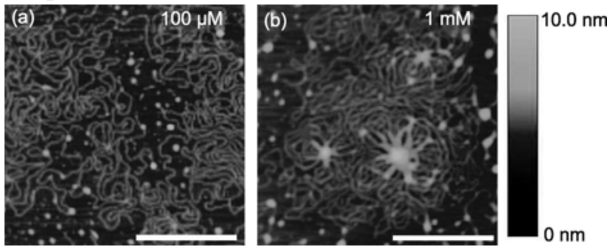

Figure 5 DNA prepared with different concentrations of spermidine on mica surfaces using method 3: pre-treating mica surfaces with spermidine for 10 min and then adding DNA to the surface. (a) 100 μM spermidine; (b) 1 mM spermidine. All scale bars are 500 nm.

(iv) Condensation on AP-mica. To further study the effect of surface modification, we carried out the condensation experiments on AP-mica, which is positively charged to have rather strong affinity for DNA molecules, and widely used as substrates for DNA adsorption in AFM experiments[19]. Except for the surface modification, the samples were prepared using method 1. As shown in Figure 6(a), untreated DNA molecules are adsorbed to the substrate due to the direct trapping by the positively charged surface, similar to the case of spermidine-treated surface (Figure 5(a)). This indicates that APTES plays a similar role as spermidine does on mica surface of inducing DNA-surface adhesion (Figure 6(a)). When pre-mixed with 100 μM spermidine, DNA exhibits a similar morphology as that in Figure 2(a). Under the condition of 400 μM spermidine, hardly any DNA could be observed (Figure 6(c)), the same as that in Fig-

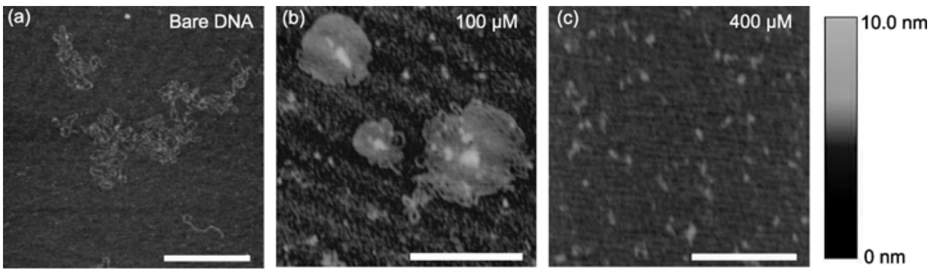

Figure 6 AFM images in air of DNA condensates on AP-mica surface. Method 1 was employed. (a) Bare DNA on AP-mica surfaces (as a control); (b) DNA patterns with 100 μM spermidine; (c) hardly any DNA can be observed when the spermidine concentration rises to 400 μM. All scale bars are 500 nm.

ure 2(e).

2.2 DNA condensation by protamine

DNA-mica adsorption versus DNA-DNA attraction. To further investigate the relationship between DNA-mica adsorption and DNA-DNA attraction, we carried out experiments on protamine-induced DNA condensation. Our experiments were based on the work in ref. [12].

As shown in Figure 7(a), when we mixed DNA (1 ng/μL) and protamine (1ng/μL) in advance, and then used 1 mM Ni^{2+} to attach the complex to mica, the pattern was flower-like, similar to that in Figure 2(a)—(d) for DNA-spermidine complex. But if we deposited DNA (in Tris-HCl buffer with 1 mM Ni^{2+}) and protamine (1 ng/μL) simultaneously onto the mica surface, the result was different (Figure 7(b)). Some parts of DNA molecules are in random coils, like DNA being adsorbed to mica by only Ni^{2+}, while some parts loosely form circular shapes (as indicted by arrow). We have noticed that although protamine contains positive charges, protamine alone cannot attach DNA to mica surfaces (pictures are not shown), so divalent cation is needed to induce DNA-mica adsorption.

Combining these phenomena, we conclude that protamine can induce DNA-DNA attraction as a result of charge neutralization, and the divalent cations provide the adsorption between DNA and the mica surface. We could observe the competition apparently between these two processes: in Figure 7(a), DNA-DNA attraction is predominant over DNA-mica adsorption so that DNA molecules exist much compactly, while in Figure 7(b), these two kinds of interaction coexist and compete, resulting in a morphology between condensation by multivalent cations (Figure 2(a)—(d)) and direct adsorption by divalent cations[20]. The possible reason may be that DNA-DNA attraction induced by protamine in such condition does not have obvious priority when compared with DNA-mica adsorption induced by Ni^{2+} which is tight for DNA binding to mica surface.

Figures 7(c) and (d) show the different DNA condensation behaviours induced by protamine on mica surfaces with different DNA adsorption strengths via Mg^{2+} concentration change. Experiments were carried out according to method 6 by adsorbing DNA to mica surface in advance, then adding protamine to the constrained DNA molecules. As a result of strong adsorption (10 mM Mg^{2+}), DNA is less condensed than the above (Figure 7(c)). However with weaker adsorption (2.5 mM Mg^{2+}), DNA may be condensed into a three-dimensional toroid (Figure 7(d)), as reported before. Therefore the condition for toroid formation on mica surface is that DNA is constrained onto mica surfaces by weak adsorption, so that DNA still has the freedom to curl at last into a toroid in the solution. When observing carefully, we find the toroid is surrounded by DNA strands, reflecting the process of toroid formation.

We also tried to condense DNA with spermidine with the same method: first attaching DNA with 2.5 mM Mg^{2+} to the surface, and then adding spermidine to the loosely pre-bound DNA as the condensing agent (pictures not shown). At 100 μM spermidine, DNA looks like random coil; at 1 mM spermidine, DNA is condensed to patterns as those shown in Figures 2(a)—(d), and no toroid was found. Comparing with Figure 2, we find that if DNA is pre-bound to the mica surface, higher concentration of spermidine is needed to make DNA condense, probably because Mg^{2+} in buffer can block the binding of spermidine to DNA; besides, it is also necessary to conquer the pre-formed adsorption between DNA and mica surface to induce DNA-DNA attraction. In higher concentration spermidine, the charges of DNA are highly neutralized, thus DNA-DNA attraction is eas-

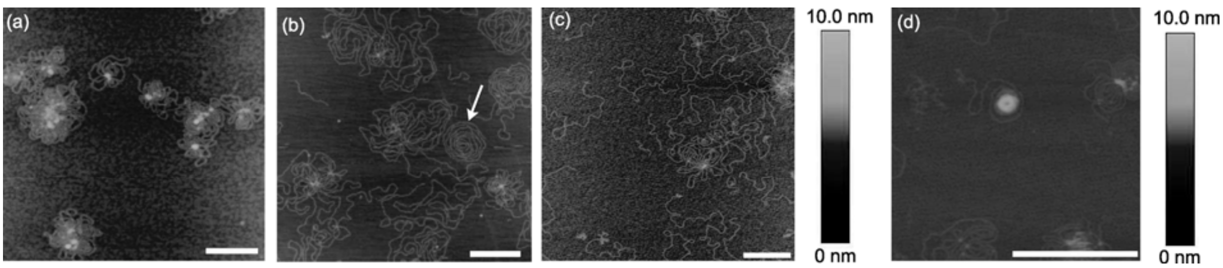

Figure 7 Morphologies of DNA condensed by protamine. (a) Pre-mixing DNA and protamine in advance, then adsorbing the complex to mica surface by 1 mM Ni^{2+} (method 4); (b) depositing DNA solution (with 1 mM Ni^{2+}) and protamine solution simultaneously onto mica surface (method 5); (c) and (d) adsorbing DNA onto mica surface with 10 and 2.5 mM Mg^{2+} first, then adding protamine to the pre-bound DNA (method 6).

ier to occur, and at last DNA is reorganized to the morphology of flat disk or flower. Different from protamine, spermidine cannot induce toroid formation in this condition. We think the reason should be that DNA-surface adsorption induced by spermidine is rather strong, thus DNA is not able to curl in three dimensions.

3 Discussion

As presented in the introduction, the flat disk-like patterns formed on mica surfaces were once recognized as the early intermediates in spermidine-induced DNA condensation whose end point was toroid, or as rearranged DNA condensate because the toroids in solution were severely disturbed by the negatively charged mica surfaces. It has been proved by several experiments that when the spermidine concentration is low, all DNA molecules exist in coiled states in the solution; some coiled DNA is transformed to compact globules when spermidine concentration is increased; and finally under high spermidine concentrations, all DNA molecules exist as compact structures. As we have observed, above 400 μM spermidine, hardly any DNA condensation forms on mica surfaces. It is thus impossible that the flat disk is the opening of the three-dimensional compact toroid formed in solution, since toroids or globules still remain the same in solution even under high spermidine concentrations[16]. The assumption of intermediates for condensates into toroids is also questionable[15], because even we increased the spermidine concentration up to 5 mM, no toroid or globule was observed. The disk or flower pattern should be two-dimensional condensations of coiled DNA in solution.

The surface-directed condensation and the spontaneous condensation in solution should be independent of and exclusive to each other. They condense under different conditions and follow different pathways. If the toroids are formed in solution in advance, two-dimensional condensations on surface will be absent. On the other hand, in the presence of spermidine, before condensing into three-dimensional toroids, the DNA molecules might be adsorbed onto mica surface preferentially, and condense into flat and circular morphologies.

In the introduction part, we referred to the work of Lin et al.[18] where toroid structures were observed with similar method. To make clear whether the difference comes from the drying method, we also carried out experiments with drying samples by heating (such as, with microwave oven or infrared lamp) instead of nitrogen stream (pictures not shown). When spermidine concentration is lower than 400 μM, globules and flat patterns coexist. When the concentration is high, only globules exist. In other words, we observed directly the coexistence of 3D (globules) and 2D (flat patterns) condensations. However toroids or ordered flat patterns were rarely observed. We think this phenomenon may be due to the fact that the heating process is not quick enough and the condensates are disturbed locally by the contracting interface between the liquid and mica surface. The heating method confirms that 3D condensations would remain on mica if not blown away by nitrogen stream. In addition, 2D pattern was not reported in ref. [18], we think the reason may be that they used water instead of Tris-HCl as buffer, so DNA might all be condensed to toroid even at low spermidine concentration.

We see that spermidine is able to reverse the charges of the mica surface, and strongly trap DNA to the surface in disordered coil morphology. However, as shown in Figure 5(b), a competition relation exists between the direct trapping by mica surface and the formation of two-dimensional condensation, or in other words, between the DNA-surface interaction and the DNA-DNA interaction. If the charge of the mica surface is reversed, and the spermidine concentration is not too high to completely screen the negative charges of DNA, DNA prefers to just bind to the mica surface as random coils without well-defined patterns (Figure 5(a)). However when the spermidine concentration is high enough to neutralize the negative charges of DNA to certain extent, the direct trap and re-organization to more ordered patterns may happen simultaneously (Figure 5(b)).

To produce two-dimensional condensations, two requirements should be satisfied: first, DNA should be attached to mica surface; second, multivalent cation is needed to neutralize the negative charges of DNA in order to form a compact structure, by reducing the repulsions between DNA strands. Two-dimensional condensation is the co-effect of DNA-mica adsorption and DNA-DNA attraction (Figure 8). As for the cores in pattern centers, how they are formed and what role it may play are still unclear.

From the formation of toroids on mica surfaces by protamine, we could understand why this kind of structures is seldom formed on mica surface by multivalent cation spermidine: adsorption between the mica surface and DNA is too strong, so as to restrict the DNA to form

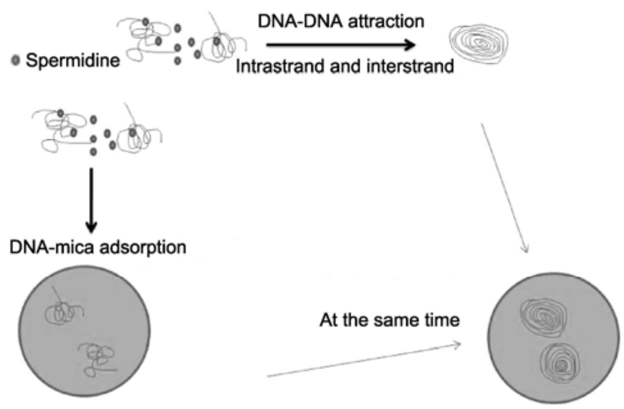

Figure 8 Flat patterns we observe on mica surfaces are due to the co-effect of DNA-mica adsorption and DNA-DNA attraction both induced by spermidine.

three-dimensional structures, instead, DNA wraps to form a rather flat and compact pattern. This further confirms that the patterns we observe in Figure 2(a)—(d) are two-dimensional wrapping of DNA on mica surfaces as a result of strong DNA-mica interaction. In the DNA-protamine experiments, protamine and divalent cation play distinct roles: inducing DNA-DNA adsorption and DNA-surface attachment, respectively[17]. However, in the DNA-spermidine experiments, these two roles are both played by spermidine itself.

We know that less compact DNA condensation is required during much of the cell life cycle, allowing protein to access DNA template to accomplish biological functions including gene translation and transcription, etc. In biological systems, there are many interfaces with which DNA can interact. Therefore it is useful to study this kind of DNA condensation on mica surfaces with structures different from that in solution. In fact, DNA condensations for two-dimensional systems have been studied experimentally. For instance, the structures of DNA molecules condensed on cationic lipid bilayers were observed to change in the presence of divalent as well as trivalent cations[21]. Besides, divalent cations could confine DNA molecules to two-dimensional cationic surfaces, inducing a much decreased DNA-DNA distance[22]. The present work should be helpful for understanding DNA behaviors on two-dimensional surfaces which should be different from those in solutions.

4 Conclusion

We have studied systematically the DNA condensations with different concentrations of multivalent cation spermidine onto negatively charged mica surface by AFM imaging. Several experimental assays have been employed. We have observed that spermidine leads DNA to form flat and single-layered patterns on mica surfaces. Their shapes are nearly circular, with a core in the centre, and DNA strands wrap around the core compactly in a relatively ordered manner. With the increase of spermidine concentration, less and less patterns can be formed and finally nothing is presented on the mica surfaces. Importantly, we have revealed that the DNA patterns like flat disks or flowers observed on mica surfaces are formed in a different pathway from which leads to DNA condensation into toroids in solutions.

Through the comparison of DNA-protamine condensation and DNA-spermidine condensation, we have found out the requirements for DNA to condense into three-dimensional toroids on surfaces, and understand why this kind of structures is difficult to be observed with spermidine.

We believe the present investigation of two-dimensional DNA condensations would be helpful for the study of interactions between biopolymers and surfaces that do exist in biological systems.

We thank Dr. Lin Zhang for helpful discussions on the observation of spermidine-induced toroids on mica surface.

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